

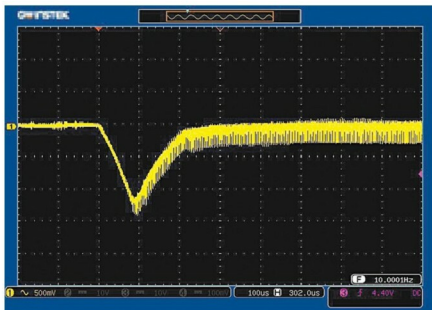


Fig. 2. Waveform chart of output voltage transition deviation of a PSU with fast voltage feedback during load power surge between 0 and 340W (scale: vertical 0.5V/div, horizontal 100µs/div)

nate them. This, in turn, will lower the weight and size of the complete power supply system.

The output voltage quality of different PSUs supplying pulse-type load can be evaluated by the waveform chart of transient output voltage deviation at step change of load power. Fig. 1 shows waveform charts of the output power of a conventional PSU (a) compared to a similar PSU with fast voltage feedback and (b) during load power transients within the range from zero to maximum. Output voltage transient deviation of the conventional unit is $+5\%/-35\%$, while output voltage transient deviation of the PSU with fast voltage feedback is $\pm 2.4\%$.

These waveform charts show that in the same conditions at load power fall and surge the transient deviation of the output voltage of different PSUs can vary approximately 15 times. That becomes possi-

ble due to very fast voltage feedback of the PSU (in less than 100µs, as shown in Fig. 2).

Such PSUs are particularly useful for transceiver units of active phased-array radars (AESAs). But

their specifications also make them a suitable product for other systems with pulse nature of load. They can be used to build fully decentralised power supply systems. Fig. 3 shows an example of such a system.

Decentralised power supply is a system where each consumer is supplied by its own relatively low-power supply unit, which, in turn, is supplied directly from the on-board mains, power generator or any other input power supply, ensuring the maximum reliability of the system as a whole. If a fuse is put at the input of each unit, in case of failure the broken cell will automatically disconnect from the input mains. Thus failure of one of the cells or its power supply will not affect other consumer operations and will not break the complete system.

Thanks to such decentralisation, communication electronics becomes invulnerable compared to a centralised system, where reliability depends on one powerful source supplying power to all consumers. Apart

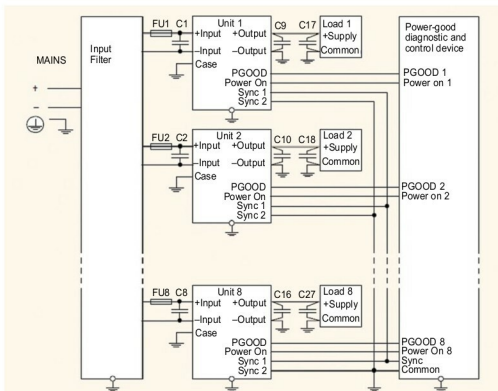


Fig. 3. Example of a decentralised power supply system with galvanically connected output voltages